TIC 258316486

Eclipsing binary candidate dossier - Star Vetter

CLEAN

```
BLS EPHEMERIS
  Period        7.522982 d
  Epoch (t0)    1572.1336 BTJD
  Duration      0.1200 d
  Depth         16.873 %  (168,730 ppm)
  SNR           160.9

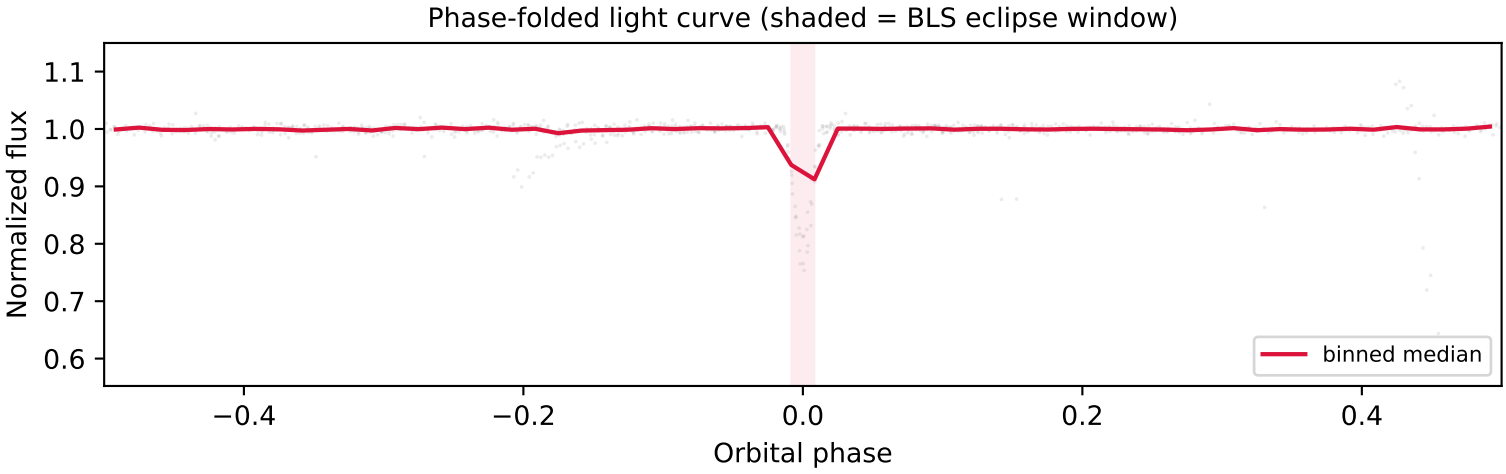
INDEPENDENT CHECKS (f3i gates)
  Eclipses observed      4  (gate >= 3)
  Robust depth ratio     1.03  (gate >= 0.50)
  Robust depth sigma     115.1  (gate >= 7.0)
  Odd/even z-score       2.98  (consistent)
  Secondary eclipse      not detected
```

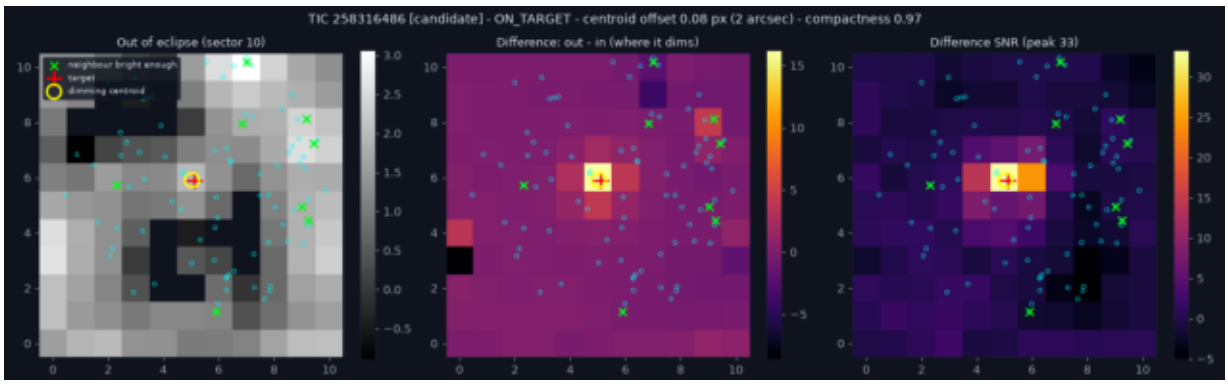
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CATALOG CROSS-MATCH (novelty)

  VSX                                  no match
  ASAS-SN                              no match
  Gaia eclipsing-binary table          no match
  Gaia variable classification         no match
  TESS EB catalog (Prsa+2022)          no match
  ExoFOP TOI list                      no match

  Verdict: NOVEL
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PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
  Sector 10    light-curve dip in this sector: 107.8 sigma    t0 refined by 0.0025 phase
  Difference-image peak SNR   33.27   (gate >= 5.0)      Compactness   0.97   (gate >= 0.50)
  Dimming centroid offset     0.08 px (2 arcsec)   (gate <= 1.0 px, i.e. on the target)
  Nearest bright-enough neighbour in cutout: TIC 258316502
    2.74 px away, dTmag +0.83 - farther from the dimming than the target
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No caveats - clean detection on every check.